

URITECT

Final Project Handoff

Release candidate 1.3.0+4 | 26 September 2026

Final Scope

Offline Android thesis prototype for markerless, semiquantitative interpretation of ten URS-10T reagent pads, with gated Bayesian UTI research interpretation and separate deterministic renal follow-up.

The 80% target is met for individual-analyte accuracy only. It is not a whole-scan or disease-diagnostic accuracy claim.

Frozen component	Version
Visual classifier	production_semiquant_knn_markerless_roi_topfix_v3_20260908
UTI research model	uti_bayesian_lr_v1_1_20260926
Renal rules	renal_followup_rules_v1.1_20260919
Android build	1.3.0+4

Engineering Completion

Area	Final state
Dataset	Laua-an and Cabatuan cleaned and ingested; 7,840 final analyte rows; zero exact duplicate rows under the frozen key.
Leakage control	Specimen-grouped 80/20 split; three lighting variants remain together.
Android inference	Ten per-analyte k-NN models and image processing execute locally in Dart; no production Internet permission.
AWB and ROI	Neutral strip/plastic gray-world normalization; markerless strip and ten-pad localization.
Invalid capture	Quality, exposure, blur, geometry, ten-pad, ambiguity, and confidence rejection paths.
UTI	v1.1 strict eligibility and grouped LRs; eligible calculations show the provisional posterior, prior, and factors without probability bands.
Renal	v1.1 rule IDs, action priority, safety override, separate results UI, and audit persistence.
Verification	49 Flutter tests pass; static analysis reports no issues.

Frozen Grouped Evaluation

Analyte	Accuracy	Macro F1	Sensitivity	Specificity	Kappa
Leukocytes	68.10%	44.50%	46.72%	85.74%	0.315
Nitrite	99.39%	83.18%	75.00%	75.00%	0.664
Urobilinogen	82.82%	64.57%	62.15%	62.15%	0.309
Protein	87.73%	25.93%	24.56%	83.34%	0.253
pH	81.60%	57.49%	52.25%	87.79%	0.357
Blood	93.25%	90.69%	88.93%	91.15%	0.714
Specific Gravity	58.90%	55.57%	56.43%	91.54%	0.494
Ketone	73.01%	67.52%	66.81%	66.81%	0.367
Bilirubin	87.12%	85.39%	84.65%	84.65%	0.708
Glucose	93.25%	37.41%	32.38%	87.75%	0.457

Aggregate: 82.52% individual-analyte accuracy (1,345/1,630); 17.18% all-ten-correct scans (28/163); 61.22% mean macro F1; 58.99% mean macro sensitivity; 81.59% mean macro specificity; 0.464 mean kappa.

At the 0.45 confidence threshold, 73.62% of scans are accepted; accepted scans achieve 83.75% individual-analyte accuracy and 19.17% all-ten-correct accuracy.

Rare semiquantitative classes have small support. Report class supports and confusion matrices with these summaries; do not use aggregate accuracy alone.

Objective Status

Objective	Status before human evaluation
1. Dataset	Engineering complete; manuscript must accurately document volunteer/control composition and provenance.
2. AWB	Implemented using neutral strip/plastic pixels; manuscript wording must match.
3. Offline ROI	Implemented markerlessly; real-device ROI overlay review remains manual.
4. k-NN and interpretation	Implemented and re-evaluated. Bayesian applies to gated UTI only; renal is rule-based.
5. ISO/IEC 25010	Instrument and APK ready; complete only after 10 medtechs test the app and results are analyzed.

Remaining Human Gates

1. Record the exact physical strip manufacturer, catalog, lot, expiry, IFU revision, reaction times, and category chart.
2. Obtain the two-medtech Bayesian parameter review.
3. Obtain final physician comparison/sign-off for renal v1.1.
4. Run target-device airplane-mode, ROI, invalid-image, latency, storage, deletion, and privacy checks.
5. Complete and analyze the 10-medtech app evaluation.
6. Apply every item in the manuscript revision guide.

Culture-confirmed prospective validation and statistical calibration review are required before calling the Bayesian output a validated clinical probability. This is a clinical-validation boundary, not an unfinished software feature.